

\*2.

Figure 2.1

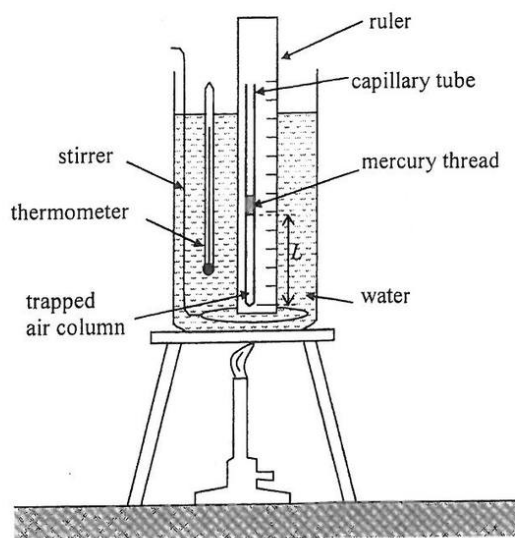


Figure 2.1 shows an air column trapped by a small mercury thread inside a uniform capillary tube. The set-up is heated by a water bath. The length of the air column  $L$  is measured at various temperature  $\theta$ . Some of the results are tabulated below:

|                                       |    |  |    |
|---------------------------------------|----|--|----|
| Temperature $\theta / ^\circ\text{C}$ | 20 |  | 92 |
| Length of air column $L / \text{mm}$  | 64 |  | 80 |

- (a) Describe the procedure(s) to be done before taking a reading in order to ensure that the trapped air reaches the same temperature as the water. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Please stick the barcode label here.

(b) Assume that length  $L$  increases linearly with temperature  $\theta$  throughout.

(i) Estimate the length of the air column when the temperature indicated by the thermometer is  $65^{\circ}\text{C}$ . (2 marks)

Answers written in the margins will not be marked.

(ii) Find the 'absolute zero' as obtained from this experiment.

(2 marks)

Answers written in the margins will not be marked.

## Book 1 section 1.1: DSE long questions

2013 Q2 (page 3)

2013-DSE-PHY 1B-5

24